

# Phase 2: Performance Tuning Report

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## Query

Our most complex query can be found in `models/core/report_customer_monthly_metrics.sql`. As the name suggests, this query determines each customer's monthly revenue, profit, orders placed, and items ordered, for each month. Of the three advanced queries, this has the most window functions and CTEs.

## Our Improvements

Table 1: Query Performance: Baseline vs Improved

| Metric                | Baseline | Improved | Changes       |
|-----------------------|----------|----------|---------------|
| Execution time        | 752ms    | 600ms    | -152ms (-20%) |
| Disk spills           | 5        | 0        | Eliminated    |
| Temp buffers written  | 4,922    | 0        | Eliminated    |
| fact_orders scan      | 134ms    | 71ms     | -63ms         |
| dim_customers scan    | 11ms     | 2ms      | -9ms          |
| Planning time         | 6.77ms   | 0.61ms   | -6ms          |
| Full buffer cache hit | No       | Yes      | Achieved      |

We implemented 3 improvements.

First, we pre-ANALYZED `fact_orders` and `dim_customers` at the start of the query:

```
ANALYZE fact_orders;  
ANALYZE dim_customers;
```

Then, we set the working memory to 128MB because we noticed that there were 5 disk spills.

```
SET work_mem = '128MB';
```

Finally, we created indexes on the `fact_orders` table, sorted by customer ID and order date, and on the `dim_customers` table, sorted by customer ID. The `INCLUDE` clause of each index makes it so that the index contains all the columns required for the query, so the original table doesn't need to be revisited.

```
CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_fact_orders_customer_date  
ON fact_orders (customer_id, order_date)  
INCLUDE (order_id, line_item_sales_total, profit)  
WHERE order_date IS NOT NULL;
```

```
CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_dim_customers_covering  
ON dim_customers (customer_id)  
INCLUDE (segment, city, state);
```

Although we thought it would be effective, the strategic indexing implemented did not improve the scan on `fact_orders` and `dim_customers` from a Seq scan to an Index scan, and we are still scanning all ~180,000 rows of the fact orders table because the full table is needed.